

GW250114 HRF

Reader guide

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Research package v72 | Proposed Zenodo release v5.3

In More Basic Terms

After two black holes merge, the resulting black hole settles down through gravitational waves, somewhat like a struck bell ringing down. This project investigates whether a proposed horizon-response model could eventually be tested with those signals.

This release improves the reliability of that test. The archived calculation closely reproduces a reference grid when specified assumptions are held fixed. But changing the released parameter row shifts the best-fitting damping rate. Reproducing the calculation therefore does not, by itself, establish a new physical effect.

The mathematics explains a limit of measurement: under the stated continuity condition, a finite experiment cannot uniformly distinguish one exact value from every value arbitrarily close to it. A meaningful test must specify a resolution, an interval procedure, or a Bayesian comparison with a declared prior.

What the methods establish

Comparison rules: checking two exclusion endpoints requires stricter control than checking one named competitor. In the stipulated Gaussian design, the effective-width screens at the historical gap are 0.100155239% and 0.115190902%, respectively. These are design requirements, not measured uncertainties for GW250114.

Archived reproducibility: the fixed-conditioning -7M calculation covers 3,400 grid points. The released-row network maximum is 200 Hz and 450 inverse seconds. A different released row gives 200 Hz and 420 inverse seconds. Several active parameters move together; the comparison does not isolate one cause.

Scientific status

The progress is a more reproducible calculation, corrected statistical comparisons, and clearer limits. It does not establish an HRF detection, a unified theory of everything, exact $\ln(4)$ or $g=4$ identification, area quantization, black-hole microstates, or a quantum horizon.

Evidence and next steps

Methods and reproducibility only | Physical interpretation remains unresolved

Fresh verification versus historical results

The September publication audit checks the supplied files and labels the checks actually rerun. Both sealed release archives match their SHA-256 sidecars; the focused manifest covers 56 files and the cumulative manifest covers 255. The package audit passes 28 checks.

The official raw strain, posterior file, and external source checkout needed for a new raw replay are not bundled. The August raw preflight, finalizer, and comparison receipts are preserved evidence from that workflow. This publication preparation does not claim that those raw computations were executed again.

A documented wording correction

The recovery audit labels a negative offset as "fresh-minus-cache." The packaged arrays instead give $\text{mean}(\text{cache} - \text{raw}) = -0.00023655252309568448$; reversing that subtraction gives a positive value. The dated publication audit records this correction. Centered residual bounds and scientific conclusions are unchanged. The sealed originals remain intact.

The next credible analysis

The proposed next-analysis protocol requires an observable-complete model, detector-aware remnant and quasinormal-mode inference, nuisance and calibration uncertainty, global target-to-alternative separation, and null calibration for the complete search. It is a planning draft, not a preregistration or an executed experiment.

Both detectors' event data have already been inspected. A new detector-disjoint reanalysis cannot retroactively make those observations blind validation data. A genuinely unexamined event or data partition would be needed for a new blind test.

Where to read more

Start with **START_HERE_V5_3_METHODS_RELEASE.md** and the **2026-09-05 publication audit**. The focused HRF v72 ZIP holds the methods, code, surfaces and receipts. The master v72 ZIP holds the cumulative research history. Read historical material through the current v72 qualifications. Use the included SHA-256 files to verify integrity.

Source basis: the supplied v72 main report, one-page certificate, recovery audit, final release verification, and sealed computational archives. Research documents retain their original July/August dates. September packaging, reader documentation and local verification were prepared with AI assistance in Codex; these checks do not constitute independent peer review.

Prior record, as documented in local history: zenodo.org/records/20192925. Proposed v5.3 must be checked against the live version history before publication. No new DOI or publication is asserted by this guide.